

Datatype mapping in MySQL and PostgreSQL

MS-SQL datatype	MySQL Equivalent	PostgreSQL Equivalent	Compatibility
INT	INTEGER	INT	Direct equivalent across all platforms
BIGINT	BIGINT	BIGINT	Same storage size and range
SMALLINT	SMALLINT	SMALLINT	Fully compatible
TINYINT	SMALLINT	TINYINT	PostgreSQL does not support unsigned tiny integers
BIT	BOOLEAN	TINYINT(1)	PostgreSQL has native BOOLEAN support; MySQL internally treats BOOLEAN as TINYINT
DECIMAL (p,s)	NUMERIC (p,s)	DECIMAL(p,s)	Preserves exact precision
NUMERIC (p,s)	NUMERIC (p,s)	NUMERIC (p,s)	Native equivalent
FLOAT	DOUBLE PRECISION	DOUBLE	Closest floating-point equivalent
REAL	REAL	FLOAT	Compatible approximate numeric type
MONEY	NUMERIC (19,4)	DECIMAL (19,4)	Prevented precision loss in financial calculations
VARCHAR(n)	VARCHAR (n)	VARCHAR (n)	Direct mapping
TEXT	TEXT	LONGTEXT	MySQL TEXT length limitations required LONGTEXT
XML	XML / TEXT	LONGTEXT	MySQL lacks native XML datatype
DATE	DATE	DATE	Native support
TIME	TIME	TIME	Native support
DATETIME	TIMESTAMP	DATETIME	PostgreSQL timestamps provide timezone handling
UNIQUE IDENTIFIER	UUID	CHAR(50)	PostgreSQL has native UUID support
BINARY(n)	BYTEA	BINARY(n)	PostgreSQL stores binary data using BYTEA
VARBINARY(n)	BYTEA	VARBINARY(n)	Preserved binary storage
IMAGE	BYTEA	LONGBLOB	IMAGE is deprecated in MSSQL

IDENTITY	SERIAL / GENERATED AS IDENTITY	AUTO_INCREMENT	Auto-generated primary keys
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